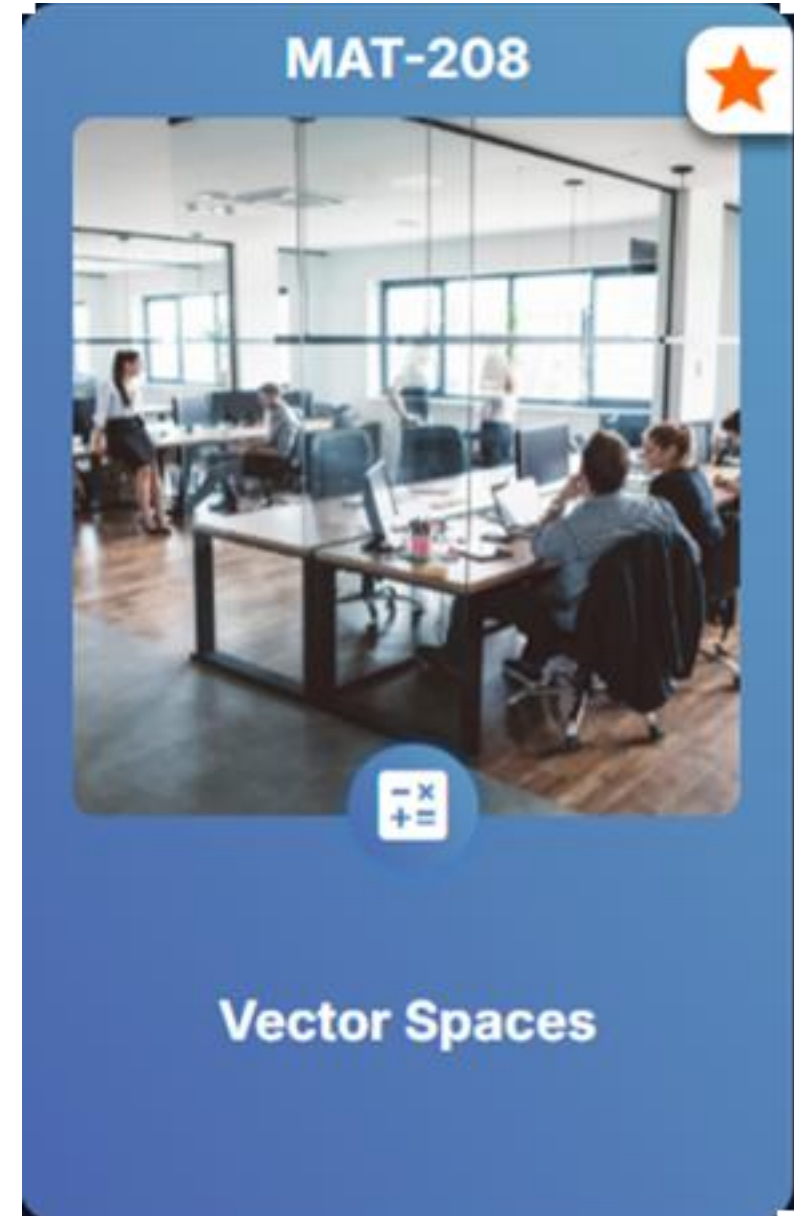


Week 2-3: Basic Vector & Matrix Operations

LEARNING BY DOING

Prof. Dr. P. Sooraksa



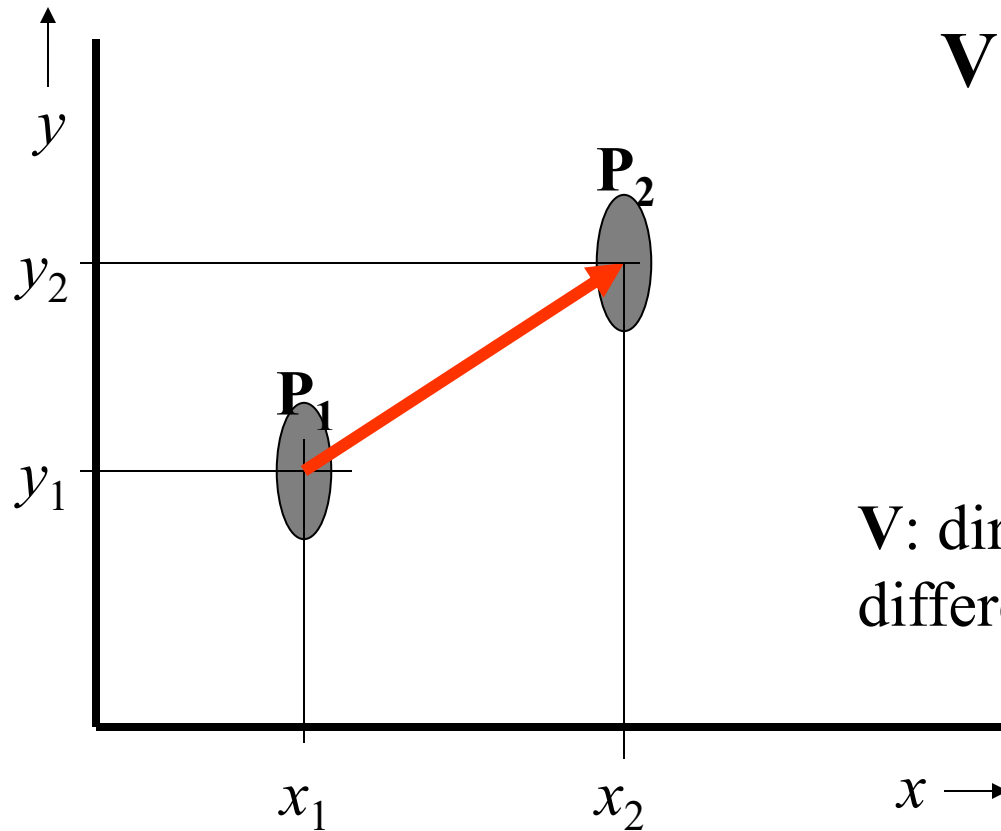
Today Topics: Fundamental Concept and Basic Operation

- Vectors & Vector Space
- Matrices and Basic Operation
- Vector and Matrix Laboratory: MATLAB

Vectors

- Vector:
 - “arrow”
 - multiple interpretations (displacement, velocity, force, ...)
 - has a magnitude and direction
 - has *no* position
- Notation: **V**, also V , **v**, v en v
- (V_x, V_y, V_z) , also (x, y, z) , (x_1, x_2, x_3)

Vectors



$$\begin{aligned}\mathbf{V} &= \mathbf{P}_2 - \mathbf{P}_1 \\ &= (x_2 - x_1, y_2 - y_1) \\ &= (V_x, V_y)\end{aligned}$$

$\mathbf{V}$: directed line segment, or
difference between two points

Vectors

Length of a vector:

$$|\mathbf{V}| = \sqrt{V_x^2 + V_y^2} \quad (2\text{D} : \text{Pythagoras})$$

$$|\mathbf{V}| = \sqrt{V_x^2 + V_y^2 + V_z^2} \quad (3\text{D})$$

Vectors

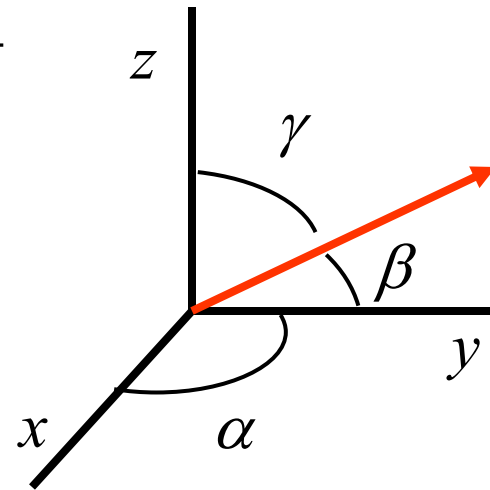
Direction of a vector: Direction angles.

$$\cos \alpha = \frac{V_x}{|\mathbf{V}|}, \cos \beta = \frac{V_y}{|\mathbf{V}|}, \cos \gamma = \frac{V_z}{|\mathbf{V}|}$$

Unit vector $\mathbf{V}$:

$$\bar{\mathbf{V}} = \frac{\mathbf{V}}{|\mathbf{V}|}$$

Magnitude info is removed, direction is kept.



Vector operations

A scalar has a magnitude (mass, time, temperature, charge).

A vector has a magnitude (its length) and a direction.

Examples: velocity, force, momentum, field strength.

Boldface letters denote vectors.

Unit vectors are denoted by $\hat{\mathbf{n}}, \hat{\mathbf{a}}$

On the blackboard I use $\vec{A}$.

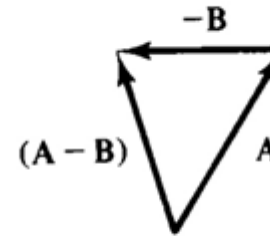
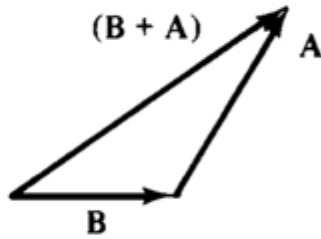
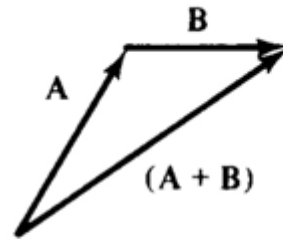




Vectors have no location.

Vector field $\mathbf{A}(\mathbf{r})$

addition of two vectors: $\mathbf{A+B}$

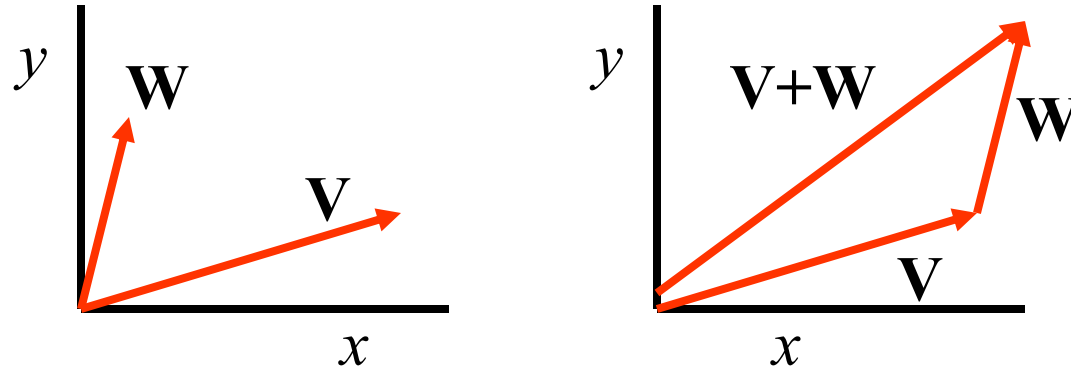


multiplication by a scalar: $a\mathbf{A}$



Vector addition

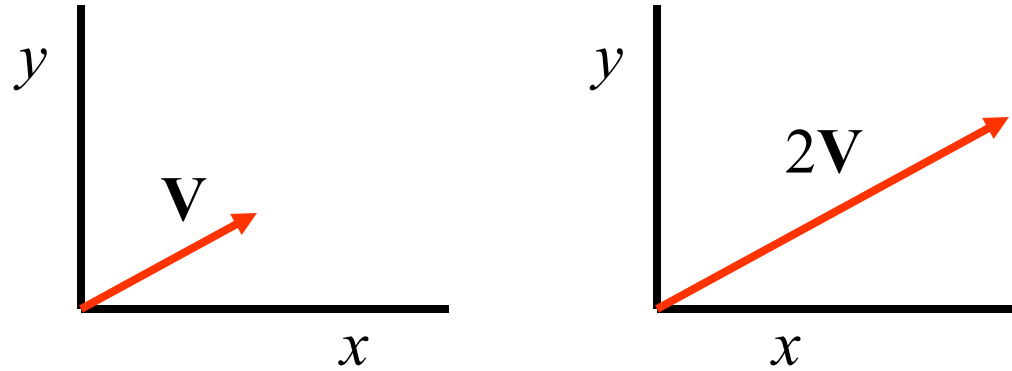
Add components, put vector head to tail



$$\mathbf{V} + \mathbf{W} = (V_x + W_x, V_y + W_y, V_z + W_z)$$

Scalar multiplication of vector

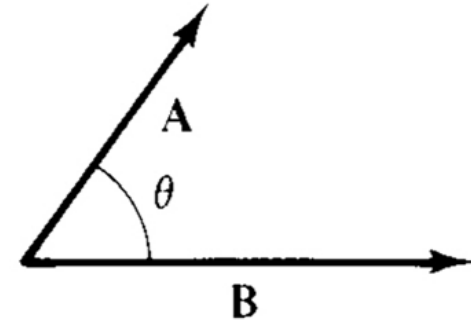
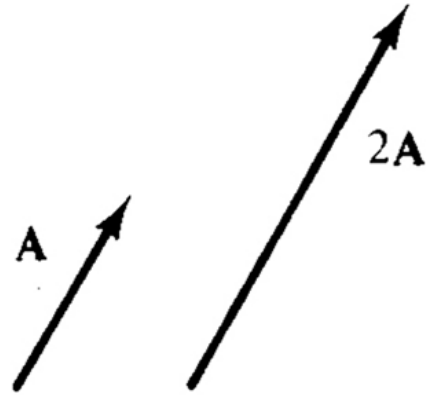
Multiplication components with scalar s



$$s\mathbf{V} = (sV_x, sV_y, sV_z)$$

dot product (scalar, inner): $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$

Example work



$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$$

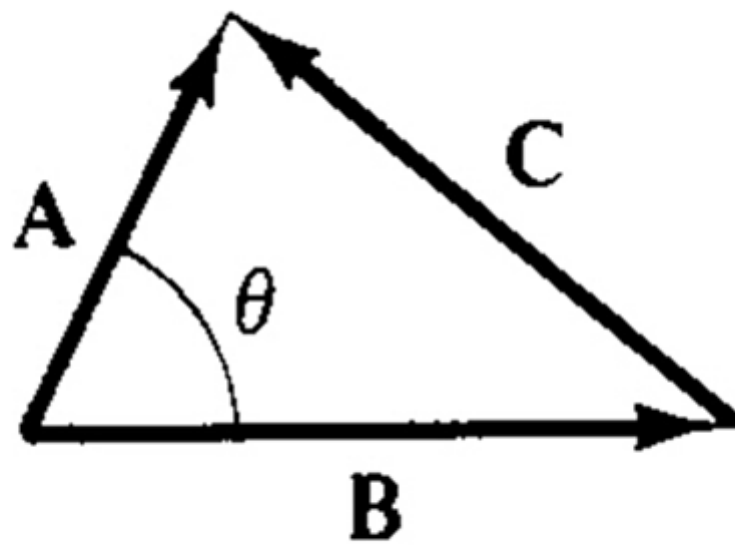
$$\mathbf{A} \cdot \mathbf{B} = AB$$

if parallel

$$\mathbf{A} \cdot \mathbf{B} = 0$$

if perpendicular

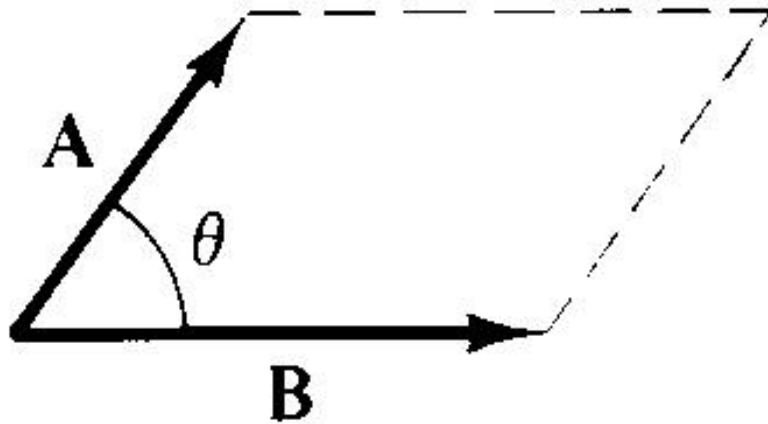
$$W = -\mathbf{F} \cdot (\mathbf{r}_2 - \mathbf{r}_1)$$



cross product (vector, outer): $\mathbf{A} \times \mathbf{B} = AB \sin \theta \hat{\mathbf{n}}$

$\hat{\mathbf{n}}$ is the unit vector perpendicular to the AB-plane.

$\mathbf{A}, \mathbf{B}, \hat{\mathbf{n}}$ form a right-handed system. $\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}$



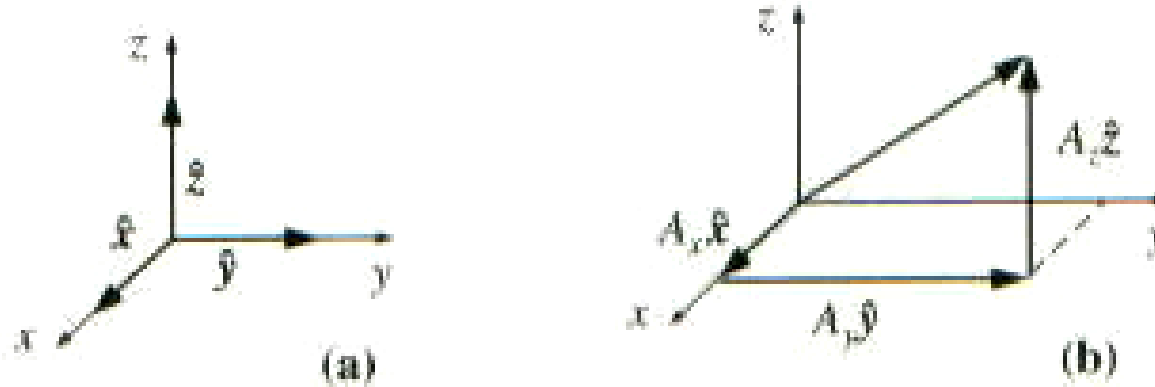
$|\mathbf{A} \times \mathbf{B}|$
is the area of the
parallelogram.

Example: angular momentum

$$\mathbf{A} \times \mathbf{A} = 0$$

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}$$

Component Form



$$\mathbf{A} = A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$$

basis: $\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$

components: A_x, A_y, A_z

$$\hat{\mathbf{n}}_i \quad A_i \quad i = 1, 2, 3 \quad 1: x, 2: y, 3: z \quad A_i = \hat{\mathbf{n}}_i \cdot \mathbf{A}$$

common notation: $\mathbf{A} = (A_x, A_y, A_z)$

Properties of the basis

$$\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j = \delta_{ij},$$

$$\delta_{ij} = 1 \quad \text{if} \quad i = j, \quad = 0 \quad \text{if} \quad i \neq j \quad \text{Kronecker symbol}$$

$$\hat{\mathbf{n}}_1 \times \hat{\mathbf{n}}_2 = \hat{\mathbf{n}}_3 \quad \text{cyclic} \quad 123 \quad 231 \quad 312$$

$$(\mathbf{A} + \mathbf{B})_i = A_i + B_i \qquad a\mathbf{A} = \sum_{i=1}^3 (aA_i) \mathbf{n}_i$$

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^3 A_i B_i$$

$$\mathbf{C} = \mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad or \quad C_i = \sum_{j,k=1}^3 \varepsilon_{ijk} A_j B_k$$

$$\begin{aligned} \varepsilon_{ijk} = +1 & \quad \text{if} \quad ijk = 123, 231, 312 \\ -1 & \quad \text{if} \quad 132, 213, 321 \\ 0 & \quad \text{else} \end{aligned}$$

Levi-Civita
symbol

Remind Determinants

Definition

The **determinant** of a 2×2 matrix A is denoted $|A|$ and is given by

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

Observe that the determinant of a 2×2 matrix is given by *the different of the products of the two diagonals* of the matrix.

The notation **det**(A) is also used for the determinant of A .

Example

$$A = \begin{bmatrix} 2 & 4 \\ -3 & 1 \end{bmatrix}$$

$$\det(A) = \begin{vmatrix} 2 & 4 \\ -3 & 1 \end{vmatrix} = (2 \times 1) - (4 \times (-3)) = 2 + 12 = 14$$

Theorem

The determinant of a square matrix is the sum of the products of the elements of any row or column and their cofactors.

$$i\text{th row expansion: } |A| = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$$

$$j\text{th column expansion: } |A| = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$$

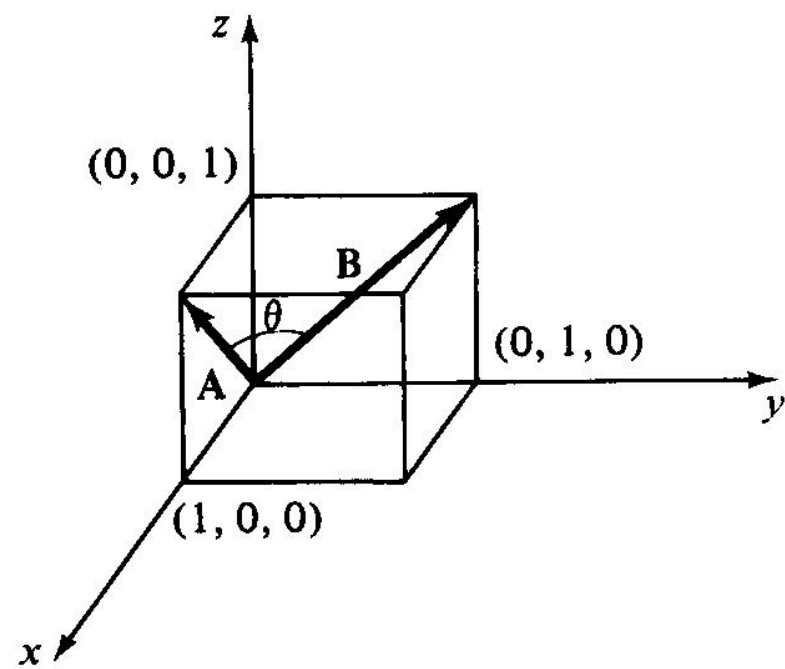
Example

Find the determinant of the following matrix using the second row.

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 1 \\ 4 & 2 & 1 \end{bmatrix}$$

Solution

$$\begin{aligned} |A| &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} \\ &= -3 \begin{vmatrix} 2 & -1 \\ 2 & 1 \end{vmatrix} + 0 \begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} \\ &= -3[(2 \times 1) - (-1 \times 2)] + 0[(1 \times 1) - (-1 \times 4)] - 1[(1 \times 2) - (2 \times 4)] \\ &= -12 + 0 + 6 = -6 \end{aligned}$$



Triple Products

scalar triple product:

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \quad \text{cyclic}$$

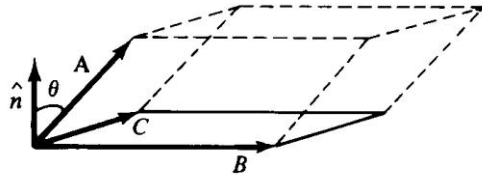


Figure 1.12

volume

$$|\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C})| = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

vector triple product:

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

bac - cab
rule

Higher order products by repeated bac-cab and symmetries of the scalar triple product.

Notation

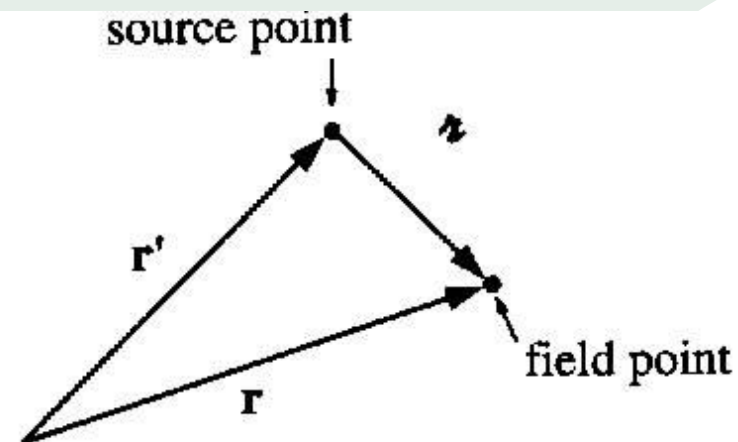
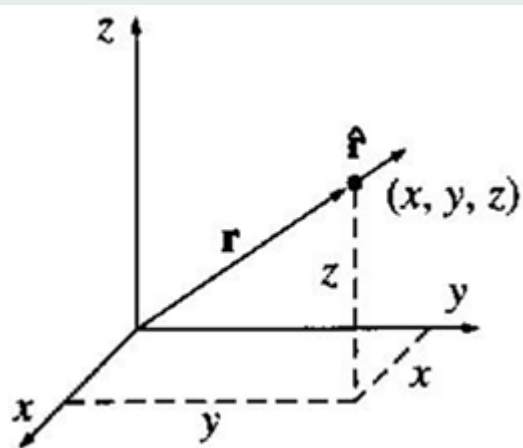
position vector:

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$$

infinitesimal displacement: $d\mathbf{l} = dx\hat{\mathbf{x}} + dy\hat{\mathbf{y}} + dz\hat{\mathbf{z}}$

separation vector:

$$\mathbf{r} = \mathbf{r} - \mathbf{r}', \quad \hat{\mathbf{r}} = \frac{\mathbf{r}}{|\mathbf{r}|}$$



Vector norms

Magnitude, modulus, or absolute value for scalars generalizes to *norm* for vectors

We will use only p -norms, defined by

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

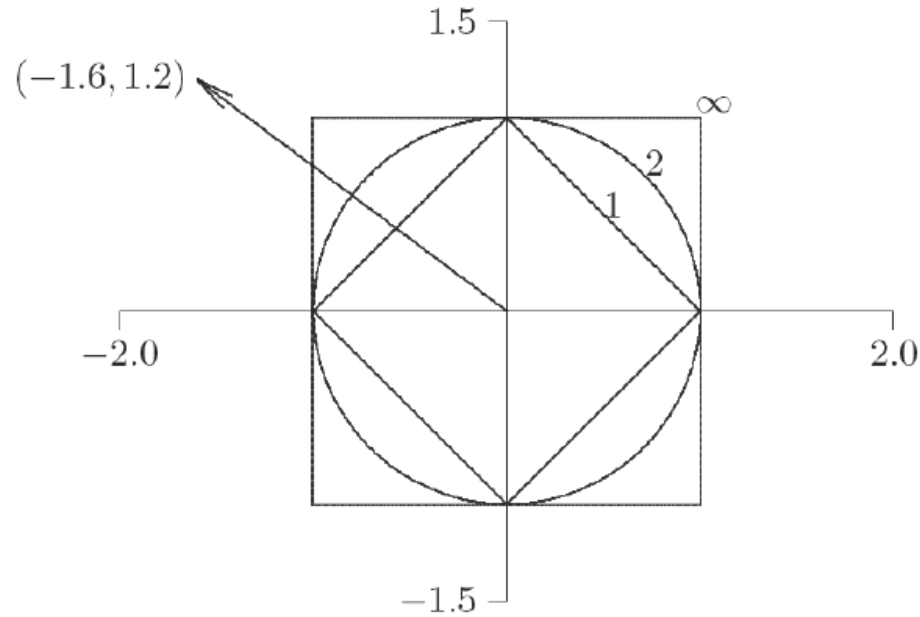
for integer $p > 0$ and n -vector $\mathbf{x}$

Important special cases

- 1-norm: $\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$
- 2-norm: $\|\mathbf{x}\|_2 = \left(\sum_{i=1}^n |x_i|^2 \right)^{1/2}$
- ∞ -norm: $\|\mathbf{x}\|_\infty = \max_i |x_i|$

Vector norms: example

Drawing shows unit sphere in two dimensions for each norm



Norms have following values for vector shown

$$\|x\|_1 = 2.8 \quad \|x\|_2 = 2.0 \quad \|x\|_\infty = 1.6$$

In general, for any vector x in $\mathbb{R}^n$, $\|x\|_1 \geq \|x\|_2 \geq \|x\|_\infty$

Vector norms: properties

For any vector norm

- $\|x\| > 0$ if $x \neq 0$
- $\|\gamma x\| = |\gamma| \cdot \|x\|$ for any scalar γ
- $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality)

In more general treatment, these properties taken as *definition* of vector norm

Useful variation on triangle inequality

- $|\|x\| - \|y\|| \leq \|x - y\|$

VECTOR SPACES

DEFINITION AND
CONCEPT

INTRODUCTION

- A **vector space** or a linear space is a group of objects called vectors, added collectively and multiplied (“scaled”) by numbers, called scalars. Scalars are usually considered to be real numbers. But there are few cases of scalar multiplication by rational numbers, complex numbers, etc. with vector spaces. The methods of vector addition and scalar multiplication must satisfy specific requirements such as axioms. Real vector space and complex vector space terms are used to define scalars as real or complex numbers.

VECTOR SPACES

- We defined a vector space as a set equipped with the binary operations of addition and scalar multiplication, a constant vector 0 , and the unary operation of negation, which satisfy several axioms. Here are the axioms again, but in abbreviated form. They hold for all vectors v and w and for all scalars c and d .

- 
- 1. $v + w = w + v$
 - 2. $(u + v) + w = u + (v + w)$
 - 3. $v + 0 = v = 0 + v$
 - 4. $v + (-v) = 0$
 - 5. $1v = v$
 - 6. $c(dv) = (cd)v$
 - 7. $c(v + w) = cv + cw$
 - 8. $(c + d)v = cv + dv$
 - There are a lot of other properties we want these operations to have, but those aren't included among the axioms because they can be proved from them.

Thus the set must follow the following properties:

PROPERTIES UNDER ADDITION:

1. closure property.
2. associative property.
3. existence of additive identity.
4. existence of additive inverse.
5. commutative property.

PROPERTIES UNDER SCALAR MULTIPLICATION:

Rest properties are the properties under multiplication.

Vector subspaces

- **(Subspace Criteria)** A subset W of a vector space V is a subspace if and only if
- The zero vector in V is in W .
- For any vectors $A, B \in W$, the addition $A+B \in W$.
- For any vector $A \in W$ and a scalar c , the scalar multiplication $cA \in W$

■ Ex: Subspace of R^2

(1) $\{\mathbf{0}\}$ $\mathbf{0} = (0, 0)$

(2) Lines through the origin

(3) R^2

■ Ex: Subspace of R^3

(1) $\{\mathbf{0}\}$ $\mathbf{0} = (0, 0, 0)$

(2) Lines through the origin

(3) Planes through the origin

(4) R^3

- Ex : (A subspace of $M_{2 \times 2}$)

Let W be the set of all 2×2 symmetric matrices. Show that

W is a subspace of the vector space $M_{2 \times 2}$, with the standard

operations of matrix addition and scalar multiplication.

Sol:

$$\because W \subseteq M_{2 \times 2} \quad M_{2 \times 2} : \text{vector sapces}$$

$$\text{Let } A_1, A_2 \in W \quad (A_1^T = A_1, A_2^T = A_2)$$

$$A_1 \in W, A_2 \in W \Rightarrow (A_1 + A_2)^T = A_1^T + A_2^T = A_1 + A_2 \quad (A_1 + A_2 \in W)$$

$$k \in R, A \in W \Rightarrow (kA)^T = kA^T = kA \quad (kA \in W)$$

$$\therefore W \text{ is a subspace of } M_{2 \times 2}$$

- Ex : (The set of singular matrices is not a subspace of $M_{2 \times 2}$)

Let W be the set of singular matrices of order 2. Show that W is not a subspace of $M_{2 \times 2}$ with the standard operations.

Sol:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in W, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in W$$

$$\therefore A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin W$$

$\therefore W_2$ is not a subspace of $M_{2 \times 2}$

- Ex : (The set of first-quadrant vectors is not a subspace of R^2)

Show that $W = \{(x_1, x_2) : x_1 \geq 0 \text{ and } x_2 \geq 0\}$, with the standard operations, is not a subspace of R^2 .

Sol:

Let $\mathbf{u} = (1, 1) \in W$

$\therefore (-1)\mathbf{u} = (-1)(1, 1) = (-1, -1) \notin W$ (not closed under scalar multiplication)

$\therefore W$ is not a subspace of R^2

Spanning Sets and Linear Independence

A vector $\mathbf{v}$ in a vector space V is called a linear combination of the vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$ in V if $\mathbf{v}$ can be written in the form

$$\mathbf{v} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots + c_k\mathbf{u}_k \quad c_1, c_2, \dots, c_k : \text{scalars}$$

Linear combination:

■ Ex : (Finding a linear combination)

$$\mathbf{v}_1 = (1,2,3) \quad \mathbf{v}_2 = (0,1,2) \quad \mathbf{v}_3 = (-1,0,1)$$

Prove (a) $\mathbf{w} = (1,1,1)$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

(b) $\mathbf{w} = (1,-2,2)$ is not a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$

Sol:

$$(a) \quad \mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

$$\begin{aligned} (1,1,1) &= c_1(1,2,3) + c_2(0,1,2) + c_3(-1,0,1) \\ &= (c_1 - c_3, 2c_1 + c_2, 3c_1 + 2c_2 + c_3) \end{aligned}$$

$$c_1 - c_3 = 1$$

$$\Rightarrow 2c_1 + c_2 = 1$$

$$3c_1 + 2c_2 + c_3 = 1$$


$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & 1 \\ 3 & 2 & 1 & 1 \end{array} \right] \xrightarrow{\text{Guass-Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow c_1 = 1 + t, \quad c_2 = -1 - 2t, \quad c_3 = t$$

(this system has infinitely many solutions)

$$\stackrel{t=1}{\Rightarrow} \mathbf{w} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3$$

(b)

$$\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 2 & 1 & 0 & -2 \\ 3 & 2 & 1 & 2 \end{array} \right] \xrightarrow{\text{Guass-Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -4 \\ 0 & 0 & 0 & 7 \end{array} \right]$$

$\Rightarrow$ this system has no solution ($\because 0 \neq 7$)

$$\Rightarrow \mathbf{w} \neq c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

- the span of a set: $\text{span}(S)$

If $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$ is a set of vectors in a vector space V , then **the span of S** is the set of all linear combinations of the vectors in S ,

$$\text{span}(S) = \{c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k \mid \forall c_i \in R\}$$

(the set of all linear combinations of vectors in S)

- a spanning set of a vector space:

If every vector in a given vector space can be written as a linear combination of vectors in a given set S , then S is called a **spanning set** of the vector space.

■ Notes:

$$\text{span}(S) = V$$

$\Rightarrow S$ spans (generates) V

V is spanned (generated) by S

S is a spanning set of V

■ Notes:

$$(1) \text{span}(\phi) = \{\mathbf{0}\}$$

$$(2) S \subseteq \text{span}(S)$$

$$(3) S_1, S_2 \subseteq V$$

$$S_1 \subseteq S_2 \Rightarrow \text{span}(S_1) \subseteq \text{span}(S_2)$$

- Linear Independent (L.I.) and Linear Dependent (L.D.):

$S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\}$: a set of vectors in a vector space V

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

- (1) If the equation has only the trivial solution ($c_1 = c_2 = \dots = c_k = 0$) then S is called linearly independent.
- (2) If the equation has a nontrivial solution (i.e., not all zeros), then S is called linearly dependent.

■ Notes:

(1) $\emptyset$ is linearly independent

(2) $\mathbf{0} \in S \Rightarrow S$ is linearly dependent.

(3) $\mathbf{v} \neq \mathbf{0} \Rightarrow \{\mathbf{v}\}$ is linearly independent

(4) $S_1 \subseteq S_2$

S_1 is linearly dependent $\Rightarrow S_2$ is linearly dependent

S_2 is linearly independent $\Rightarrow S_1$ is linearly independent

■ Ex : (Testing for linearly independent)

Determine whether the following set of vectors in R^3 is L.I. or L.D.

$$S = \{(1, 2, 3), (0, 1, 2), (-2, 0, 1)\}$$

Sol:

$$\begin{array}{ccccccc} & \mathbf{v}_1 & & \mathbf{v}_2 & & \mathbf{v}_3 & \\ & & & & c_1 & & -2c_3 = 0 \\ c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0} & \Rightarrow & 2c_1 + & c_2 + & & = 0 \\ & & & & 3c_1 + 2c_2 + & c_3 = 0 \end{array}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right] \xrightarrow{\text{Gauss - Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\Rightarrow c_1 = c_2 = c_3 = 0 \quad (\text{only the trivial solution})$$

$$\Rightarrow S \text{ is linearly independent}$$

■ Ex : (Testing for linearly independent)

Determine whether the following set of vectors in P_2 is L.I. or L.D.

$$S = \{ \underset{\mathbf{v}_1}{1+x-2x^2}, \underset{\mathbf{v}_2}{2+5x-x^2}, \underset{\mathbf{v}_3}{x+x^2} \}$$

Sol:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = \mathbf{0}$$

$$\text{i.e. } c_1(1+x-2x^2) + c_2(2+5x-x^2) + c_3(x+x^2) = 0+0x+0x^2$$

$$\Rightarrow \begin{cases} c_1 + 2c_2 = 0 \\ c_1 + 5c_2 + c_3 = 0 \\ -2c_1 - c_2 + c_3 = 0 \end{cases} \Rightarrow \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 1 & 5 & 1 & 0 \\ -2 & -1 & 1 & 0 \end{array} \right] \xrightarrow{\text{G.J.}} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\Rightarrow$ This system has infinitely many solutions.

(i.e., This system has nontrivial solutions.)

$\Rightarrow S$ is linearly dependent. (Ex: $c_1=2$, $c_2=-1$, $c_3=3$)

■ Ex : (Testing for linearly independent)

Determine whether the following set of vectors in 2×2

matrix space is L.I. or L.D.

$$S = \left\{ \underset{\mathbf{v}_1}{\begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}}, \underset{\mathbf{v}_2}{\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix}}, \underset{\mathbf{v}_3}{\begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix}} \right\}$$

Sol:

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{0}$$

$$c_1 \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{aligned} 2c_1 + 3c_2 + c_3 &= 0 \\ c_1 &= 0 \\ 2c_2 + 2c_3 &= 0 \\ c_1 + c_2 &= 0 \end{aligned}$$

$$\Rightarrow \left[\begin{array}{ccc|c} 2 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 2 & 2 & 0 \\ 1 & 1 & 0 & 0 \end{array} \right] \xrightarrow{\text{Gauss - Jordan Elimination}} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow c_1 = c_2 = c_3 = 0 \text{ (This system has only the trivial solution.)}$$

$$\Rightarrow S \text{ is linearly independent.}$$

What is a matrix?

Matrix:

- Mathematical objects with operations

Matrix in computer graphics:

- Defines a coordinate frame
- Defines a transformation
- Handy tool for manipulating transformations

Matrix

Matrix: rectangular array of *elements*

Element: quantity (value, expression, function, ...)

Examples:

$$\begin{bmatrix} 3.60 & -0.01 & 2.1 \\ -5.46 & 0.00 & 1.6 \end{bmatrix}, \begin{bmatrix} e^x & x \\ e^{2x} & x^2 \end{bmatrix}, [a_1 \quad a_2 \quad a_3], \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Matrix

$r \times c$ matrix: r rows, c columns

m_{ij} : element at row i and column j .

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1c} \\ m_{21} & m_{22} & \cdots & m_{2c} \\ \vdots & \vdots & & \vdots \\ m_{r1} & m_{r2} & \cdots & m_{rc} \end{bmatrix}$$

Matrix

$r \times c$ matrix: r rows, c columns

m_{ij} : element at row i and column j .

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1c} \\ m_{21} & m_{22} & \cdots & m_{2c} \\ \vdots & \vdots & & \vdots \\ m_{r1} & m_{r2} & \cdots & m_{rc} \end{bmatrix}$$

$r \times c$
elements

Matrix

$r \times c$ matrix: r rows, c columns

m_{ij} : element at row i and column j .

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1c} \\ m_{21} & m_{22} & \cdots & m_{2c} \\ \vdots & \vdots & & \vdots \\ m_{r1} & m_{r2} & \cdots & m_{rc} \end{bmatrix}$$

c columns

Matrix

$r \times c$ matrix: r rows, c columns

m_{ij} : element at row i and column j .

$$\mathbf{M} = \begin{bmatrix} m_{11} & m_{12} & \cdots & m_{1c} \\ m_{21} & m_{22} & \cdots & m_{2c} \\ \vdots & \vdots & & \vdots \\ m_{r1} & m_{r2} & \cdots & m_{rc} \end{bmatrix} \quad r \text{ rows}$$

Matrix

Column vector:
matrix with $c=1$

$$\mathbf{V} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_r \end{bmatrix}$$

Used for vectors
(and points)

Row vector:
matrix with $r=1$

$$\mathbf{V} = [v_1 \quad v_2 \quad \cdots \quad v_c]$$

Scalar:
matrix with $r=1$ and $c=1$

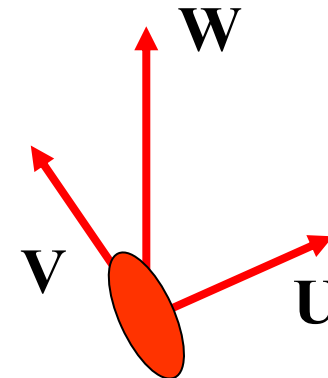
$$\mathbf{V} = [v]$$

Matrix

Matrix as collection of column vectors:

$$\mathbf{M} = [\mathbf{U} \quad \mathbf{V} \quad \mathbf{W}] = \begin{bmatrix} u_x & v_x & w_x \\ u_y & v_y & w_y \\ u_z & v_z & w_z \end{bmatrix}$$

Matrix contains an axis-frame:



Operations on matrices

- Multiplication with scalar
 - simple
- Addition
 - simple
- Matrix-matrix multiplication
 - More difficult, but the most important

Scalar Matrix multiplication

Matrix **M** multiplied with scalar s :

$$\mathbf{A} = s\mathbf{M} = \begin{bmatrix} sm_{11} & sm_{12} & \cdots & sm_{1c} \\ sm_{21} & sm_{22} & \cdots & sm_{2c} \\ \vdots & \vdots & & \vdots \\ sm_{r1} & sm_{r2} & \cdots & sm_{rc} \end{bmatrix}, \text{ i.e., } a_{ij} = sm_{ij}$$

$$\text{example: } \begin{bmatrix} 3x & 12 \\ 6 & 2y \end{bmatrix} = 3 \begin{bmatrix} x & 4 \\ 2 & y \end{bmatrix}$$

Scalar Matrix addition

$$\begin{aligned}\mathbf{M} = \mathbf{A} + \mathbf{B} &= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ a_{21} & a_{22} & \cdots & a_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1c} \\ b_{21} & b_{22} & \cdots & b_{2c} \\ \vdots & \vdots & & \vdots \\ b_{r1} & b_{r2} & \cdots & b_{rc} \end{bmatrix} \\ &= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1c} + b_{1c} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2c} + b_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} + b_{r1} & a_{r2} + b_{r2} & \cdots & a_{rc} + b_{rc} \end{bmatrix}, \text{i.e., } m_{ij} = a_{ij} + b_{ij}\end{aligned}$$

$$\text{example: } \begin{bmatrix} 3x & 12 \\ 6 & 2y \end{bmatrix} + \begin{bmatrix} 4 & -10 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 3x+4 & 2 \\ 8 & 2y \end{bmatrix}$$

Scalar Matrix addition

$$\mathbf{M} = \mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ a_{21} & a_{22} & \cdots & a_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1c} \\ b_{21} & b_{22} & \cdots & b_{2c} \\ \vdots & \vdots & & \vdots \\ b_{r1} & b_{r2} & \cdots & b_{rc} \end{bmatrix}$$
$$= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1c} + b_{1c} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2c} + b_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} + b_{r1} & a_{r2} + b_{r2} & \cdots & a_{rc} + b_{rc} \end{bmatrix}, \text{ i.e., } m_{ij} = a_{ij} + b_{ij}$$

- Just add elements pairwise
- A and B must have the same number of rows and columns
- Generalization of vector and scalar addition

Matrix Matrix multiplication

$$\mathbf{C} = \mathbf{AB} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \boxed{a_{21} & a_{22} & \cdots & a_{2n}} \\ \vdots & \vdots & & \vdots \\ a_{mn} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} \boxed{b_{11}} & b_{12} & \cdots & b_{1q} \\ \boxed{b_{21}} & b_{22} & \cdots & b_{2q} \\ \vdots & \vdots & & \vdots \\ \boxed{b_{p1}} & b_{p2} & \cdots & b_{pq} \end{bmatrix}$$

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj} \quad \begin{matrix} i = 2 ; k \rightarrow & j = 1 ; k \downarrow \end{matrix}$$

- c_{ij} : dot product of **row vector** a_i and **column vector** b_j
- #columns $\mathbf{A}$ must be the same as #rows $\mathbf{B}$: $n = p$
- $\mathbf{C}$: $m \times q$ matrix

Matrix Matrix multiplication

Example:

$$\begin{bmatrix} 2 & 3 \\ -1 & -4 \\ \boxed{0} & \boxed{5} \end{bmatrix} \begin{bmatrix} 1 \\ \boxed{2} \\ 3 \\ \boxed{4} \end{bmatrix} = \begin{bmatrix} 2 \times 1 + 3 \times 3 & 2 \times 2 + 3 \times 4 \\ -1 \times 1 - 4 \times 3 & -1 \times 2 - 4 \times 4 \\ 0 \times 1 + 5 \times 3 & \boxed{0 \times 2 + 5 \times 4} \end{bmatrix} = \begin{bmatrix} 11 & 16 \\ -13 & -16 \\ 15 & \boxed{20} \end{bmatrix}$$

$i = 3$ $j = 2$ $i=3, j=2$

Matrix Matrix multiplication

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\mathbf{AB} = \dots$$

$$\mathbf{BA} = \dots$$

Matrix Matrix multiplication

Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \times 4 + 2 \times 5 + 3 \times 6 \end{bmatrix} = \begin{bmatrix} 32 \end{bmatrix}$$

$$\mathbf{BA} = \dots$$

Matrix Matrix multiplication

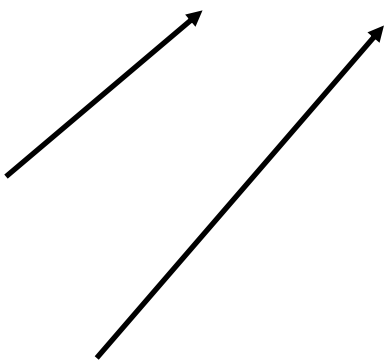
Example:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \mathbf{B} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \times 4 + 2 \times 5 + 3 \times 6 \end{bmatrix} = \begin{bmatrix} 32 \end{bmatrix}$$

$$\mathbf{BA} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 4 \times 1 & 4 \times 2 & 4 \times 3 \\ 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 6 \times 1 & 6 \times 2 & 6 \times 3 \end{bmatrix} = \begin{bmatrix} 4 & 8 & 12 \\ 5 & 10 & 15 \\ 6 & 12 & 18 \end{bmatrix}$$

AB $\neq$ BA



Matrix Matrix multiplication

- **$\mathbf{AB} \neq \mathbf{BA}$**

Matrix matrix multiplication is not commutative!

Order matters!

- **$\mathbf{A}(\mathbf{B}+\mathbf{C}) = \mathbf{AB}+\mathbf{AC}$**

Matrix matrix multiplication is distributive

Example transformation sequence (coordinate version)

$$X' \leftarrow X : \quad x' = px + qy, \quad y' = rx + sy;$$

$$X'' \leftarrow X' : \quad x'' = ax' + by', \quad y'' = cx' + dy'.$$

Then $X'' \leftarrow X :$

$$x'' = a(px + qy) + b(rx + sy),$$

$$y'' = c(px + qy) + d(rx + sy);$$

or

$$x'' = (ap + br)x + (aq + bs)y,$$

$$y'' = (cp + dr)x + (cq + ds)y.$$

Unclear!

Error-prone!

Example transformation sequence (matrix vector version)

$$X' \leftarrow X : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix};$$

$$X'' \leftarrow X' : \begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x' \\ y' \end{bmatrix}.$$

Then $X'' \leftarrow X$:

$$\begin{bmatrix} x'' \\ y'' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Example transformation sequence (compact matrix vector version)

$$X' \leftarrow X : \mathbf{X}' = \mathbf{A}\mathbf{X};$$

$$X'' \leftarrow X' : \mathbf{X}'' = \mathbf{B}\mathbf{X}'$$

$$\text{Then } X'' \leftarrow X :$$

$$\mathbf{X}'' = \mathbf{B}\mathbf{A}\mathbf{X}$$

Here some more detail:

$$\mathbf{X} = \begin{bmatrix} x \\ y \end{bmatrix}, \mathbf{X}' = \begin{bmatrix} x' \\ y' \end{bmatrix}, \mathbf{X}'' = \begin{bmatrix} x'' \\ y'' \end{bmatrix},$$

$$\mathbf{A} = \begin{bmatrix} p & q \\ r & s \end{bmatrix}; \mathbf{B} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Matrix vector notation allows for compactness
and genericity!

Matrix Transpose

$$\mathbf{M} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1c} \\ a_{21} & a_{22} & \cdots & a_{2c} \\ \vdots & \vdots & & \vdots \\ a_{r1} & a_{r2} & \cdots & a_{rc} \end{bmatrix}$$

$$\mathbf{M}^T = \begin{bmatrix} a_{11} & a_{21} & \cdots & a_{r1} \\ a_{12} & a_{22} & \cdots & a_{r2} \\ \vdots & \vdots & & \vdots \\ a_{1c} & a_{2c} & \cdots & a_{rc} \end{bmatrix}, \text{ i.e., } m_{ji}^T = m_{ij}$$

Transpose matrix:

Interchange rows and columns.

$r \times c$ matrix $\mathbf{M} \rightarrow c \times r$ matrix $\mathbf{M}^T$

examples: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}, \quad [x \ y \ z]^T = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

Matrix Inverse

Simple algebra puzzle. Let $ax = b$. What is x ?

$$\begin{array}{ll} ax = b & \text{Multiply left and right with } a^{-1} \\ a^{-1}ax = a^{-1}b & \text{Use } a^{-1}a \text{ equals } 1 \\ x = a^{-1}b & \text{Done} \end{array}$$

Linear algebra puzzle. Let $\mathbf{MU} = \mathbf{V}$. What is $\mathbf{U}$?

$$\begin{array}{ll} \mathbf{MU} = \mathbf{V} & \text{Multiply left and right with } \mathbf{M}^{-1} \\ \mathbf{M}^{-1}\mathbf{MU} = \mathbf{M}^{-1}\mathbf{V} & \text{Use } \mathbf{M}^{-1}\mathbf{M} \text{ equals } \mathbf{I} \\ \mathbf{U} = \mathbf{M}^{-1}\mathbf{V} & \text{Done!} \end{array}$$

Matrix Inverse

Inverse of a $n \times n$ (square) matrix $\mathbf{M}$ is denoted $\mathbf{M}^{-1}$, with

$$\mathbf{M}\mathbf{M}^{-1} = \mathbf{M}^{-1}\mathbf{M} = \mathbf{I},$$

where $\mathbf{I}$ is the identity matrix :

$$\mathbf{I} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}, \text{ i.e., } \mathbf{I}_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}.$$

$\mathbf{I}$ is a *diagonal matrix*, with all 1's on the diagonal.

$\mathbf{M}^{-1}$ "undoes" the effect of $\mathbf{M}$,
multiplying with $\mathbf{I}$ has no effect.

Matrix Inverse examples

$$[4]^{-1} = [1/4]$$

$$\begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix}^{-1} = \begin{bmatrix} 2 & -1 \\ -5/2 & 3/2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 5 & 4 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -5/2 & 3/2 \end{bmatrix} = \begin{bmatrix} 3 \times 2 - 2 \times 5/2 & -3 \times 1 + 2 \times 3/2 \\ 5 \times 2 - 4 \times 5/2 & -5 \times 1 + 4 \times 3/2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Matrix Inverse examples

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dots?$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & -ab + ba \\ -ab + ba & -cb + ad \end{bmatrix} = (ad - bc) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}^{-1} ? \quad \text{Does not exist, cannot be inverted.}$$

Matrix Inverse

- Does not always exist
- In general: if *determinant* = $|\mathbf{M}| = 0$, matrix cannot be inverted
- Inverse for $n = 1$ and $n = 2$: easy
- Inverse for higher n : use library function
- Important special case: *orthonormal matrix*.

Questions

If

$$A = \begin{pmatrix} 3 & 1 & 4 \\ -2 & 0 & 1 \\ 1 & 2 & 2 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 & 0 & 2 \\ -3 & 1 & 1 \\ 2 & -4 & 1 \end{pmatrix}$$

compute (a) $2A$ (b) $A + B$ (c) $2A - 3B$ (d) $(2A)^T - (3B)^T$ (e) AB (f) BA
(g) $A^T B^T$ (h) $(BA)^T$

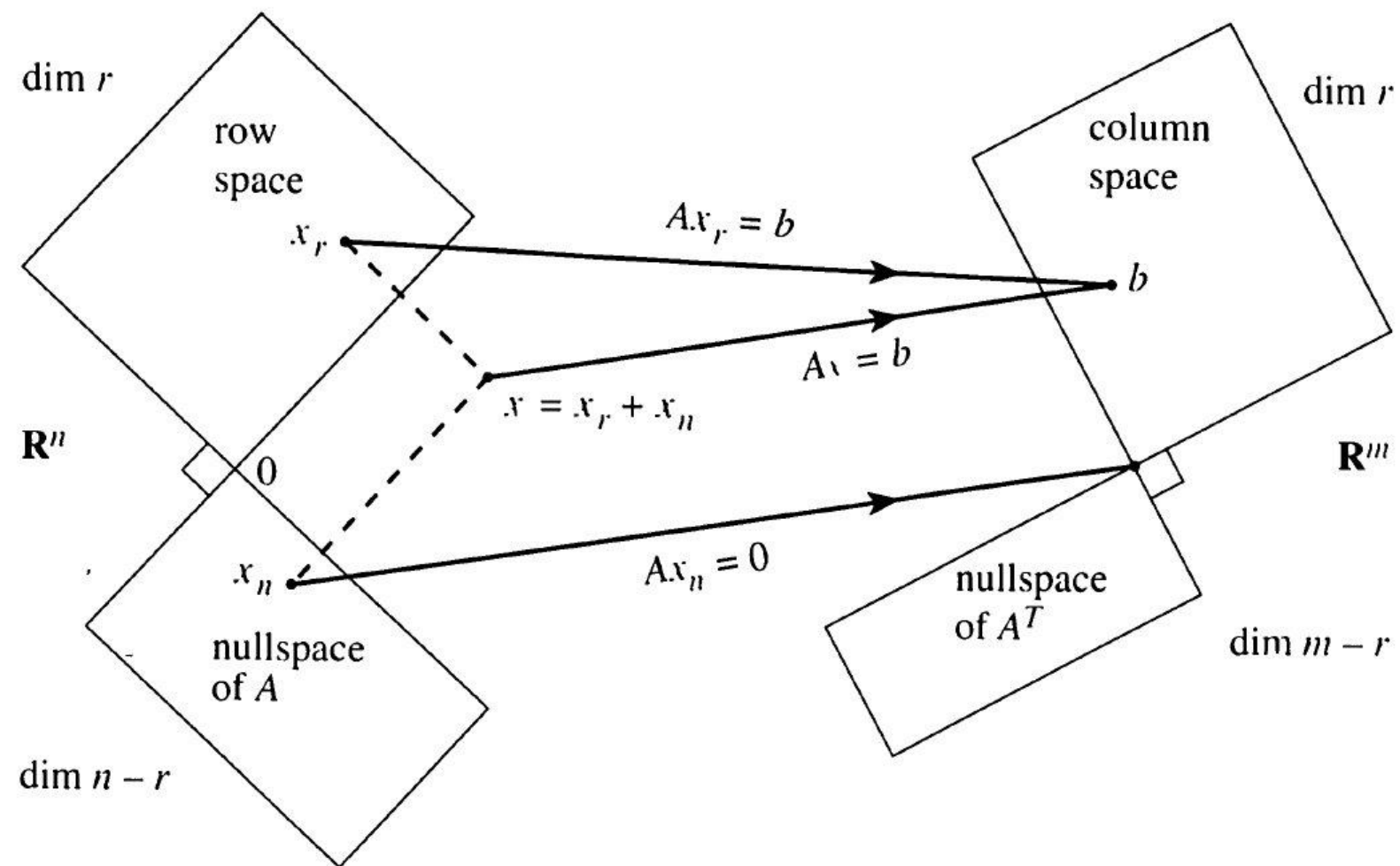
Note:

Properties of matrix transposition

Let A and B be $m \times n$ matrices and α a real number.

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(\alpha A)^T = \alpha A^T$

Note:



Questions

Given

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

show that R is nonsingular and $R^{-1} = R^T$.

Note:

Orthogonal Matrix

A Square matrix 'A' is orthogonal if

$$A^T = A^{-1}$$

(OR)

$$AA^T = A^T A = I, \text{ where}$$

- A^T = Transpose of A
- A^{-1} = Inverse of A
- I = Identity matrix of same order as 'A'

To **rotate** 45° about the origin, we apply the matrix

$$\begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} = \frac{\sqrt{2}}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

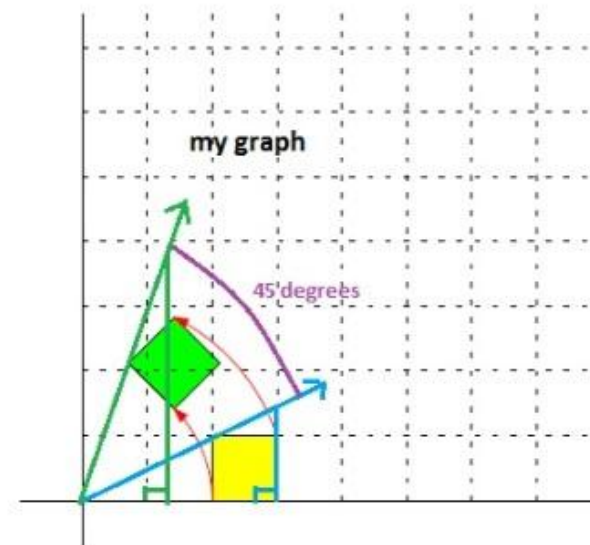
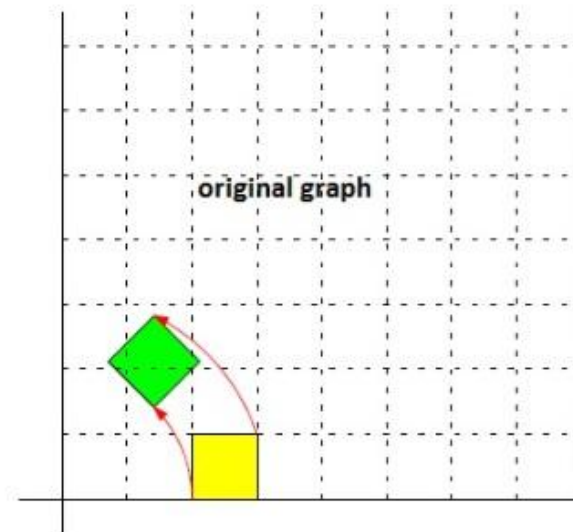
Note: $\frac{\sqrt{2}}{2} = \cos 45^\circ = \sin 45^\circ$, so this is the same as

$$\begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix}$$

Note:

Counter Clockwise $\begin{pmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{pmatrix}$

Clockwise $\begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix}$



Questions

Let A and B be $n \times n$ matrices and let $C = AB$. Prove that if B is singular then C must be singular.

The **skew symmetric** matrix

- Also called anti symmetric $\mathbf{S}^T = -\mathbf{S}$
 $\mathbf{S}^T + \mathbf{S} = 0$
- Always singular $\det \mathbf{S} = 0$

Note:

Real Case	Complex Case
$u \cdot v = u^T v = v^T u$	$u \cdot v = v^* u$
Transpose $()^T$	Conjugate transpose $()^*$
Orthogonal matrix $AA^T = I$	Unitary matrix $UU^* = I$
Symmetric matrix $A = A^T$	Hermitian matrix $H = H^*$

Note:

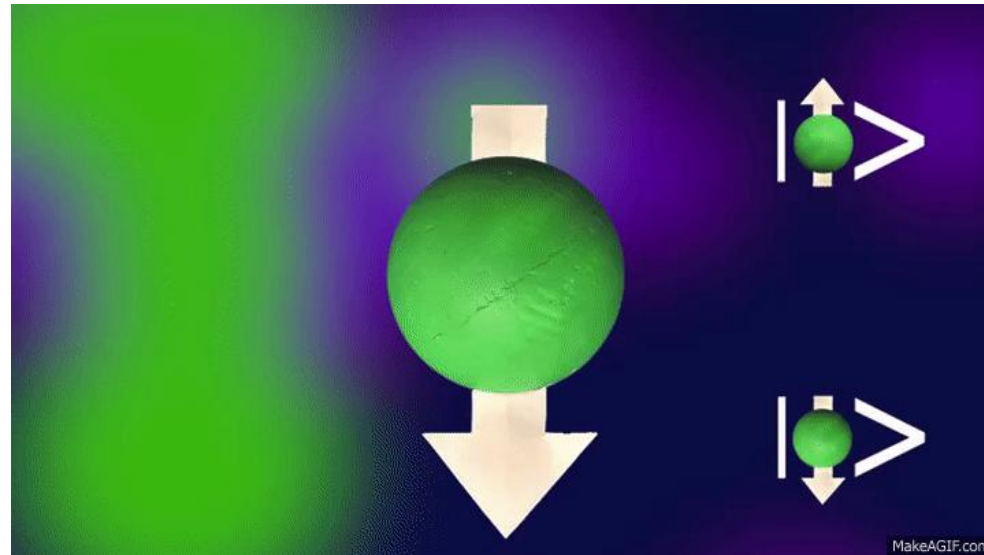
Is this matrix a Hermitian matrix?

$$\mathbf{H} = \begin{bmatrix} 1 + 3i & 5i \\ 4 & 6 + i \end{bmatrix}$$
$$\mathbf{H}^\dagger = \begin{bmatrix} 1 - 3i & 4 \\ -5i & 6 - i \end{bmatrix}$$

Pauli matrices:

Specific Hermitian matrices used in Quantum Mechanics and Quantum Computing

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

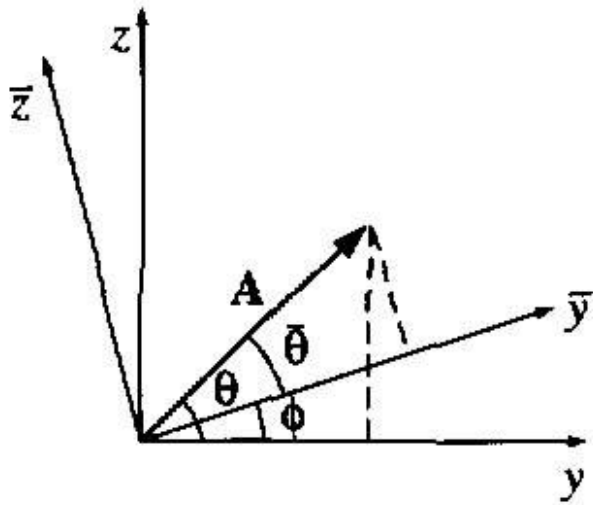


Vectors & Matrices Connection

- Vector = special case of matrix ($n \times 1$)
- Matrices transform vectors (rotation, scaling, reflection)
- Linear systems: $A \cdot x = b$

How Vectors Transform

Rotation about the x-axis:



$$\begin{pmatrix} \bar{A}_y \\ \bar{A}_z \end{pmatrix} = \begin{pmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{pmatrix} \begin{pmatrix} A_y \\ A_z \end{pmatrix}$$

In general

$$\bar{A}_i = \sum_{j=1}^3 R_{ij} A_j \quad \mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^3 A_i B_i = \sum_{i=1}^3 \bar{A}_i \bar{B}_i$$

Matrix norm

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

Norm of matrix measures maximum stretching matrix does to any vector in given vector norm

Matrix norm corresponding to vector 1-norm is maximum absolute *column* sum

$$\|A\|_1 = \max_j \sum_{i=1}^n |a_{ij}|$$

Matrix norm corresponding to vector ∞ -norm is maximum absolute *row* sum

$$\|A\|_\infty = \max_i \sum_{j=1}^n |a_{ij}|$$

Handy way to remember these is that matrix norms agree with corresponding vector norms for $n \times 1$ matrix

Matrix norm properties

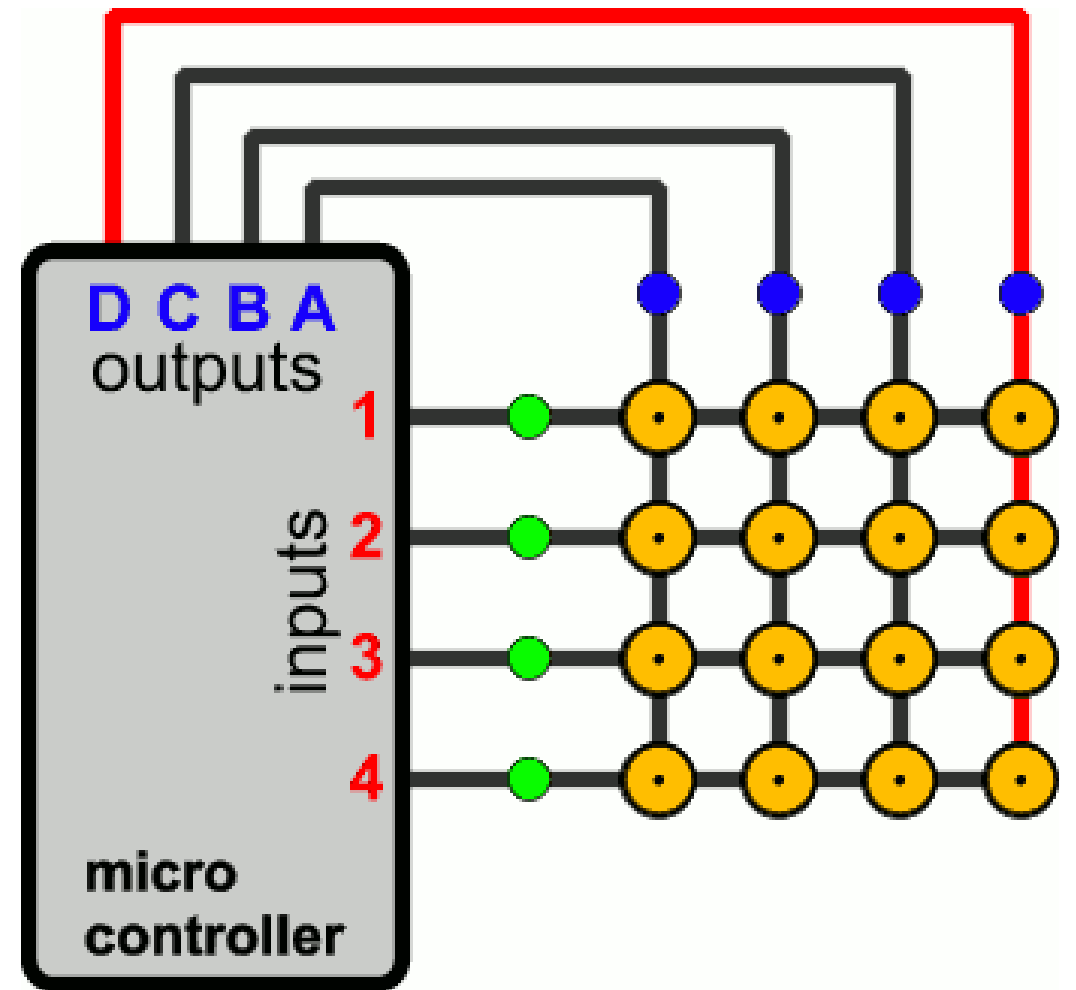
Any matrix norm satisfies

- $\|A\| > 0$ if $A \neq 0$
- $\|\gamma A\| = |\gamma| \cdot \|A\|$ for any scalar γ
- $\|A + B\| \leq \|A\| + \|B\|$

Matrix norms we have defined also satisfy

- $\|AB\| \leq \|A\| \cdot \|B\|$
- $\|Ax\| \leq \|A\| \cdot \|x\|$ for any vector x

VECTOR and MATRIX LAB



MATLAB

- MATLAB is a program for doing numerical computation. It was originally designed for solving linear algebra type problems using matrices. It's name is derived from

MATrix LABoratory.

- MATLAB has since been expanded and now has built-in functions for solving problems requiring data analysis, signal processing, optimization, and several other types of scientific computations. It also contains functions for 2-D and 3-D graphics and animation.

MATLAB

- When you first open MATLAB, notice:
 1. The **command window** is where you'll give MATLAB its input and view its output.
 2. The **workspace** shows you all of your current working variables and other objects.
 3. The **history** shows you all commands you used in CW.
 4. The **Editor** for MATLAB scripts (M-files) . To save & run the m-file type 'F5'. To open the editor with a new or old m-file use the command **open** *file_name*

MATLAB help

- For help, command description etc use F1 or following commands:
 - `help command_name`
 - `helpwin command_name`
 - `doc command_name`
 - `helpdesk command_name`
 - `demo command_name`
 - `lookfor keyword` (search unknown command)
- <http://www.mathworks.com/support/>
- For example when running “[help sin](#)” one get
SIN Sine of argument in radians.
SIN(X) is the sine of the elements of X.
See also ASIN, SIND.
Overloaded functions
Reference page in Help browser doc sin

Some Useful commands

- **what** List all m-files in current directory
- **dir/ls** List all files in current directory
- **type t** Display test.m in command window
- **delete test** Delete test.m
- **cd/chdir** Change directory
- **pwd** Show current directory
- **which test** Display directory path to 'closest' test.m
- **who** List known variables
- **whos** List known variables plus their size
- **clear** Clear variables from workspace
- **clc** Clear the command window

MATLAB & Matrices

- MATLAB treats all variables as matrices. For our purposes a matrix can be thought of as an array, in fact, that is how it is stored.
- Vectors are special forms of matrices and contain only one row OR one column.
- Scalars are matrices with only one row AND one column

Variable Names

- Variable names **ARE** case sensitive
- Variable names can contain up to 63 characters (as of MATLAB 6.5 and newer). One can use **namelengthmax** command to verify it.
- Variable names must start with a letter followed by letters, digits, and underscores.
- MATLAB variables are defined by assignment. There is no need to declare in advance the variables that we want to use or their type.
- Example

x=1;	% Define the scalar variable x
y=[1 2 3]	% row vector
z=[1;2;3]	% column vector
A=[1 2 3;4 5 6;7 8 9]	% 3x3 matrix
whos	% List of the variables defined
- Note: terminate statement with semicolon (;) to suppress output.

Special Variables

Default variable name for results

Value of π

Smallest incremental number

Infinity

Not a number e.g. 0/0

imaginary unit i , i.e. square root of -1

realmin

The smallest usable positive realnumber

realmax

The largest usable positive real number

[SpecialVars.m](#)

Other symbols

>>	prompt
...	continue statement on next line
,	separate statements and data
%	start comment which ends at end of line
;	(1) suppress output (2) used as a row separator in a matrix
:	specify range

Relational Operators

- MATLAB supports six relational operators.

Less Than	<
Less Than or Equal	<=
Greater Than	>
Greater Than or Equal	>=
Equal To	==
Not Equal To	~=

Math & Assignment Operators

Power	$\wedge$	or	$\cdot \wedge$	$a \wedge b$	or	$a \cdot \wedge b$
Multiplication	$*$	or	$\cdot *$	$a * b$	or	$a \cdot * b$
Division	$/$	or	$\cdot /$	a / b	or	$a \cdot / b$
or	$\backslash$	or	$\cdot \backslash$	$b \backslash a$	or	$b \cdot \backslash a$

NOTE: $56 / 8 = 8 \backslash 56$

– (unary) + (unary)						
Addition	$+$		$a + b$			
Subtraction	$-$		$a - b$			
Assignment	$=$		$a = b$	(assign b to a)		

[Operators.m](#)

MATLAB Logical Operators

- MATLAB supports five logical operators.

element wise/scalar logical NOT

element wise logical AND

element wise logical OR

logical (short-circuit) AND

logical (short-circuit) AND

Logical Functions

- MATLAB also supports some logical functions.
 - **xor** (a, b) exclusive or
 - **any**(x) returns 1 if any element of x is nonzero
 - **all**(x) returns 1 if all elements of x are nonzero
 - **isnan**(x) returns 1 at each NaN in x
 - **isinf**(x) returns 1 at each infinity in x
 - **finite**(x) returns 1 at each finite value in x
 - **find**(x) find indices and values of non zero elements

Some Matrix functions

- **zeros**(rows, cols) – create zero matrix
- **rand**(rows, cols) – generate random matrix
- **ones**(rows, cols) – matrix with 1 in all entries
- **eye** (rows, cols) – identity matrix
- **sub2ind, ind2sub** indices manipulation

Extracting a Sub-Matrix

- A portion of a matrix can be extracted and stored in a smaller matrix by specifying the names of both matrices and the rows and columns to extract. The syntax is:

```
sub_matrix = matrix ( r1 : rn , c1 : cn ) ;
```

```
sub_matrix = matrix ( r1 : rn , : ) ;
```

```
sub_matrix = matrix ( : , c1 : cn ) ;
```

```
sub_matrix = matrix ( r1 : dr : rn , c1 : dc : cn ) ;
```

where **r1** and **rn** specify the beginning and ending rows and **c1** and **cn** specify the beginning and ending columns to be extracted to make the new matrix.

The terms **dr** and **dc** define spacing different than one.

- [matrix.m](#)

“Continuous” functions

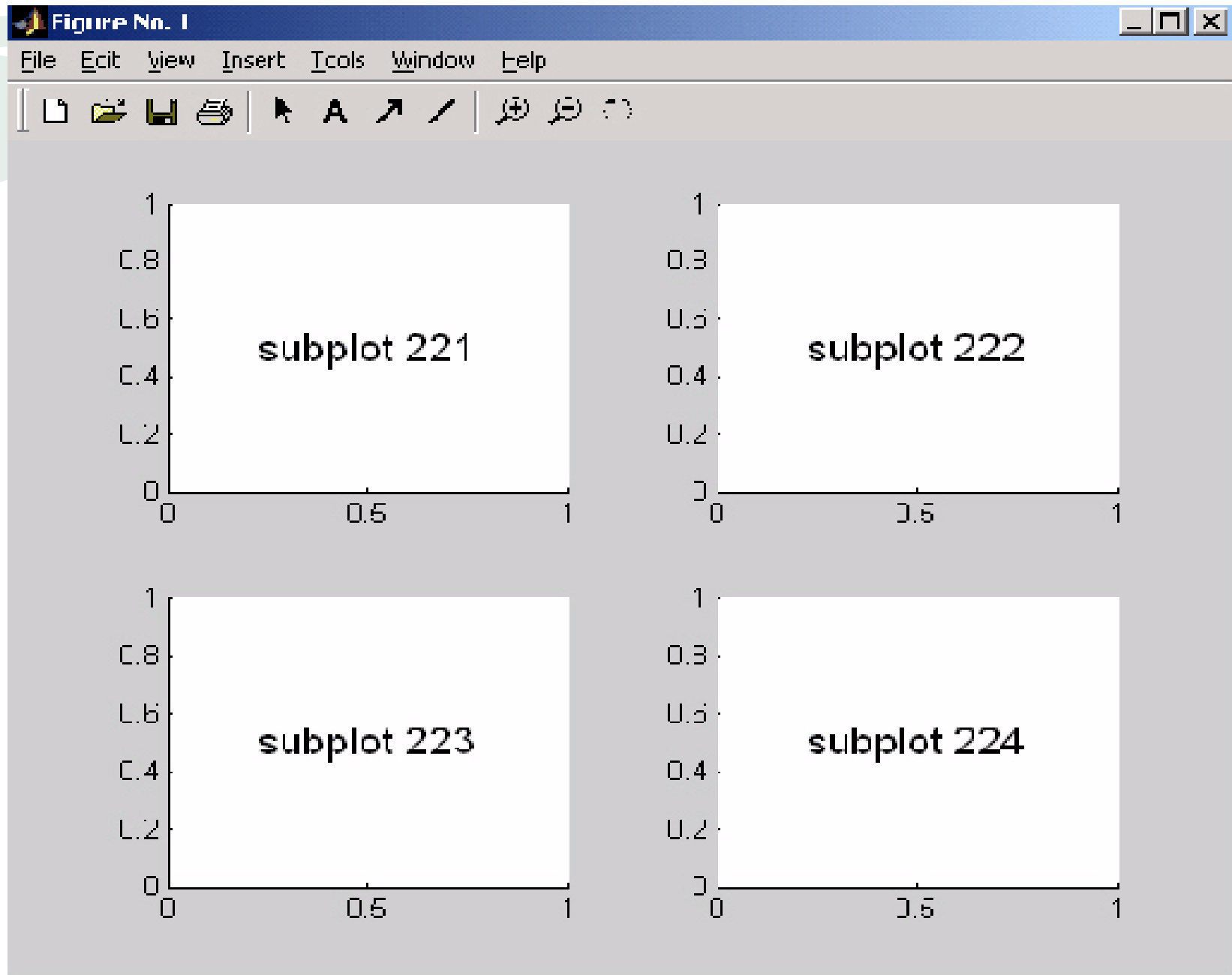
- Numerically, we cannot represent a general continuous function $(x, f(x))$ because it requires handling infinite data (for each point in the range, we need to keep $f(x)$). Instead, we represent a function by its values at a finite number of data points $(x_i, f(x_i))$, where the series of points $\{x_i\}$ is typically referred to as the sampling points or the grid points. Accordingly, the "continuous" functions in Matlab accept a vector of point $\{x_i\}$ and return a vector of values $\{f(x_i)\}$.
- Some functions
 - sqrt
 - log
 - cos/acos/sin/asin etc
 - exp - exponential
 - abs
 - sign
 - norm
 - sum
 - prod - product

Plotting

- MATLAB will plot one vector vs. another. The first one will be treated as the abscissa (or x) vector and the second as the ordinate (or y) vector. The vectors have to be the same length.
- MATLAB will also plot a vector vs. its own index. The index will be treated as the abscissa vector. Given a vector "time" and a vector "dist" we could say:

```
>> plot (time, dist)           % plotting versus time  
>> plot (time + i*dist)       % plotting versus time  
>> plot (dist)                 % plotting versus index
```

- Sometime we want to see it with different color\line stile
- And sometime we want to plot few functions in graphs
>> plot(...), hold, plot(...)
- To split page to several axes check use
>> subplot (rows, cols, place)



Plotting

- There are commands in MATLAB to "annotate" a plot to put on axis labels, titles, and legends. For example:
- To put a label on the axes we would use:
>> xlabel ('X-axis label')
>> ylabel ('Y-axis label')
- To put a title on the plot, we would use:
>> title ('Title of my plot')
- To distinct between function in the graph use:
>> legend(legend_1, legend_2)
- [plotting.m](#)

Flow control (condition)

- An if - elseif - else structure. (Note that elseif is one word)
 if expression1
 statements1
 elseif expression2
 statements2
 else
 statements3
 end
- An switch-case structure
 switch switch_expr
 case case_expr
 statement, ..., statement
 case {case_expr1, case_expr2, case_expr3, ...}
 statement, ..., statement
 otherwise
 statement, ..., statement
 end

Flow control (loops)

- A for loop in MATLAB
for *ind* = 1:100
 b(ind)=sin(ind/10)
end
- Alternative (Most of the loops can be avoided!!!):
x=0.1:0.1:10;
b=sin(x);
- A while loop in
while x <= 10
 % execute these commands
end

M-Files

- An **M-file** might be used as a **script**, i.e. file consist set of statements
- In additional, one use M-files to write **function**, in this case the file starts with function definition like:

function y = f(x)

function [u,v] = f(x,y,z)

- **File name** and the **name of function** in the file are usually **identical**, however while they are different, MATLAB use file name to call function.
- If you add additional function in same M-file, it considered sub-function and might be called from inside the M-file only. Only the first function might be called from outside.

Saving Results

- We can save all our results for future reference .
- The command
`diary 'FileName'`
saves all output to command window into the FileName.txt file
until this option is turned off by the command
`diary off`
- The following commands save & load the entire workspace into the file 'MyMatFile.mat'
 - `save 'MyMatFile'`
 - `load 'MyMatFile'`
 - `save 'x.mat' x` % save a specific variable
- saving in ASCII format:
 - `x = (-1:0.4:1)';     y = sin(x*pi)`
 - `var = [x y]` % double-column
 - `save 'my_sin.dat' -ASCII -double var` %Save in 16-digit ASCII format

Vectors and Matrices

- In MATLAB a vector can be defined as row vector or as a column vector.
- A vector of length n can be visualized as matrix of size $1 \times n$ or $n \times 1$.

```
>> x=[1 6 3 5]
```

```
x =
```

```
    1    6    3    5
```

```
>> whos
```

Name	Size	Bytes	Class
x	1x4	32	double array
y	4x1	32	double array

```
>> y=[2; 5; 7; 8]
```

```
y =
```

```
    2  
    5  
    7  
    8
```

Vectors and Matrices

- Matrices can be of different sizes, either square matrices of size $N \times N$ or rectangular $N \times M$.
- Note that vectors are special case of matrices.

```
>>  
>> C=[1 2 3;4 5 6]
```

```
C =
```

```
    1    2    3  
    4    5    6
```

```
>> A=[7 2 4; 63 8 3; 9 8 1]
```

```
A =
```

```
     7     2     4  
    63     8     3  
     9     8     1
```

```
>> B=[7 2 4;
```

```
    63 8 3;
```

```
    9 8 1]
```

```
B =
```

```
     7     2     4  
    63     8     3  
     9     8     1
```

```
>> whos
```

Name	Size	Bytes	Class
A	3x3	72	double array
B	3x3	72	double array
C	2x3	48	double array

Some Special Vectors and Matrices

```
>>  
>> ov=ones(1, 4)
```

```
ov =
```

```
      1      1      1      1
```

```
>>  
>> om=ones(3,3)
```

```
om =
```

```
      1      1      1  
      1      1      1  
      1      1      1
```

```
>>  
>> zv=zeros(1, 4)
```

```
zv =
```

```
      0      0      0      0
```

```
>>  
>> zm=zeros(3,3)
```

```
zm =
```

```
      0      0      0  
      0      0      0  
      0      0      0
```

Matrix Calculations

- Addition

```
>> A=eye(3,3)
```

A =

```
    1    0    0
    0    1    0
    0    0    1
```

```
>> B=ones(3,3)
```

B =

```
    1    1    1
    1    1    1
    1    1    1
```

```
>> C=A+B
```

C =

```
    2    1    1
    1    2    1
    1    1    2
```

- $\wedge$

```
>> A=[1 2;3 4]
```

A =

```
    1    2
    3    4
```

```
>> B=[1 1; 2 2]
```

B =

```
    1    1
    2    2
```

```
>> C=A*B
```

C =

```
    5    5
   11   11
```

- $|r$

```
>> A=[1 3 1; 1 2 1; 1 5 0]
```

A =

```
    1    3    1
    1    2    1
    1    5    0
```

```
>> det(A)
```

ans =

```
    1
```

```
>> A^(-1)
```

ans =

```
   -5    5    1
    1   -1    0
    3   -2   -1
```

```
>> inv(A)
```

ans =

```
   -5    5    1
    1   -1    0
    3   -2   -1
```

Solving Simultaneous Equations

$Ax=b$

- A=matrix of coefficients,
- x = vector of variables,
- b = vector of values.

$$x_1 + 2x_2 = 1$$

$$2x_1 + 3x_2 = 2$$

- In matrix form the above system and its solution can be written as

$$Ax = b \quad x = A^{-1}b$$

- In MATLAB we write as $x = \text{inv}(A)*b$

```
>> A=[1 2;2 3];    >> inv(A)*b
>> b=[1;2];        ans =
>>
                        1
                        0
```